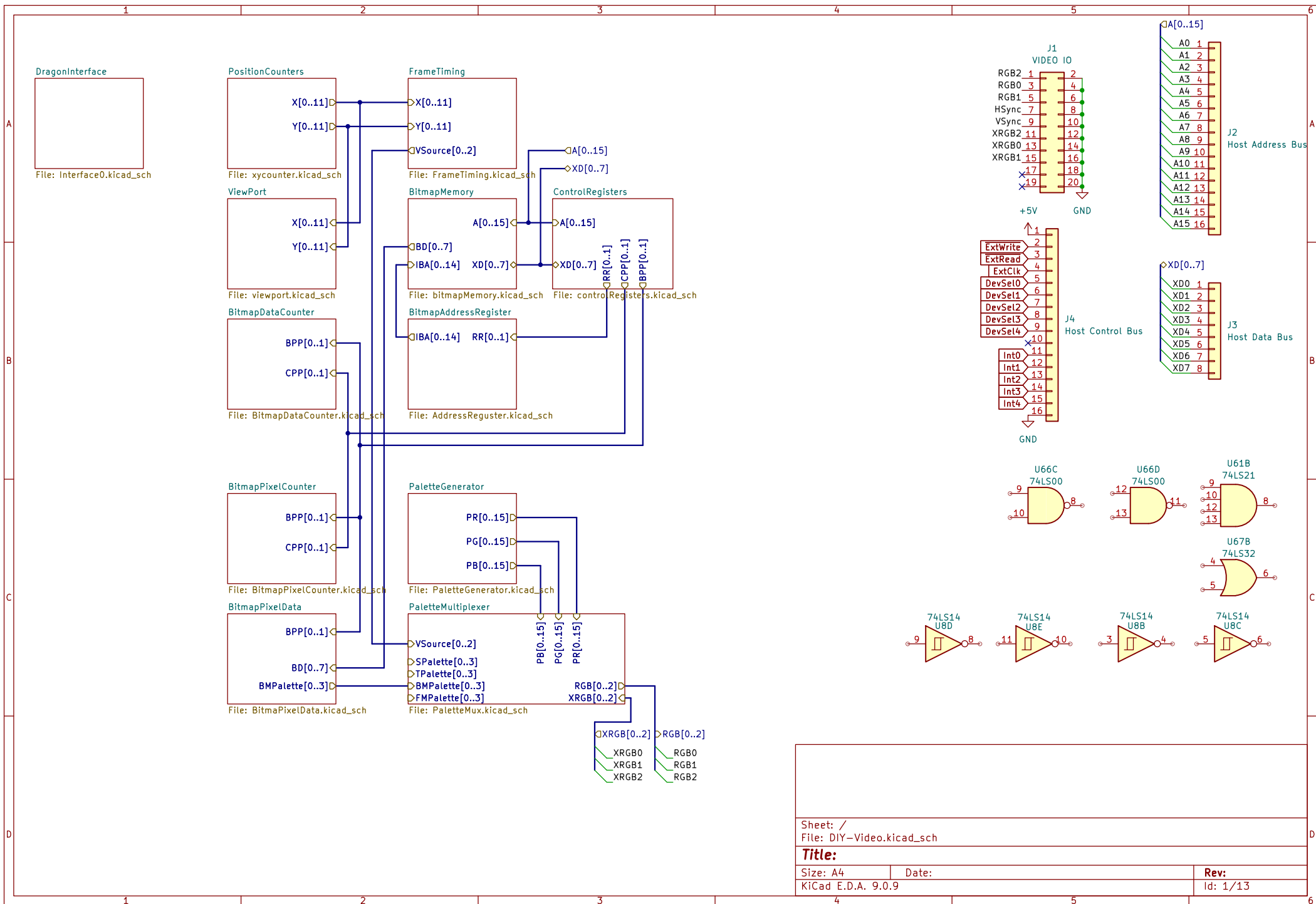
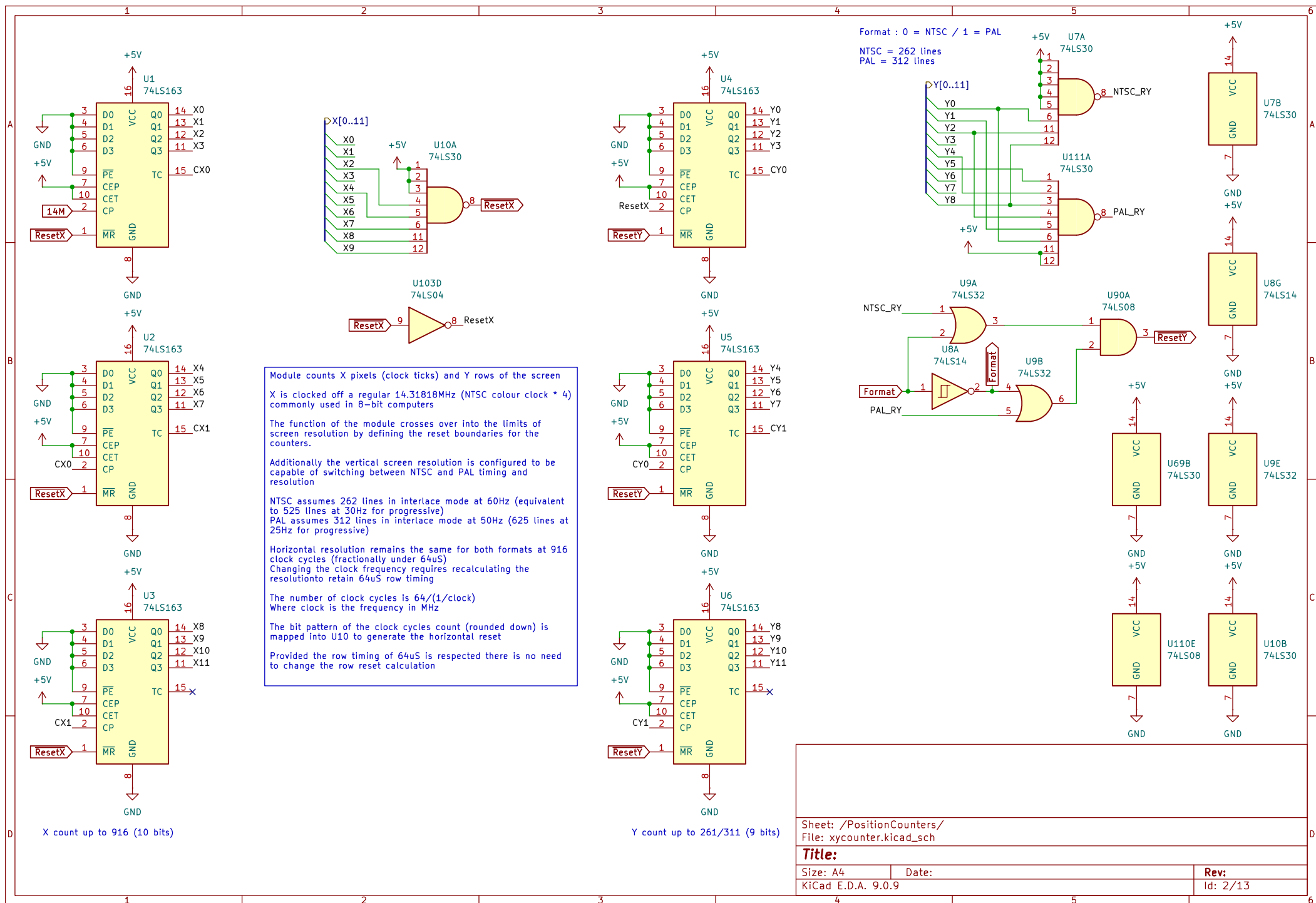
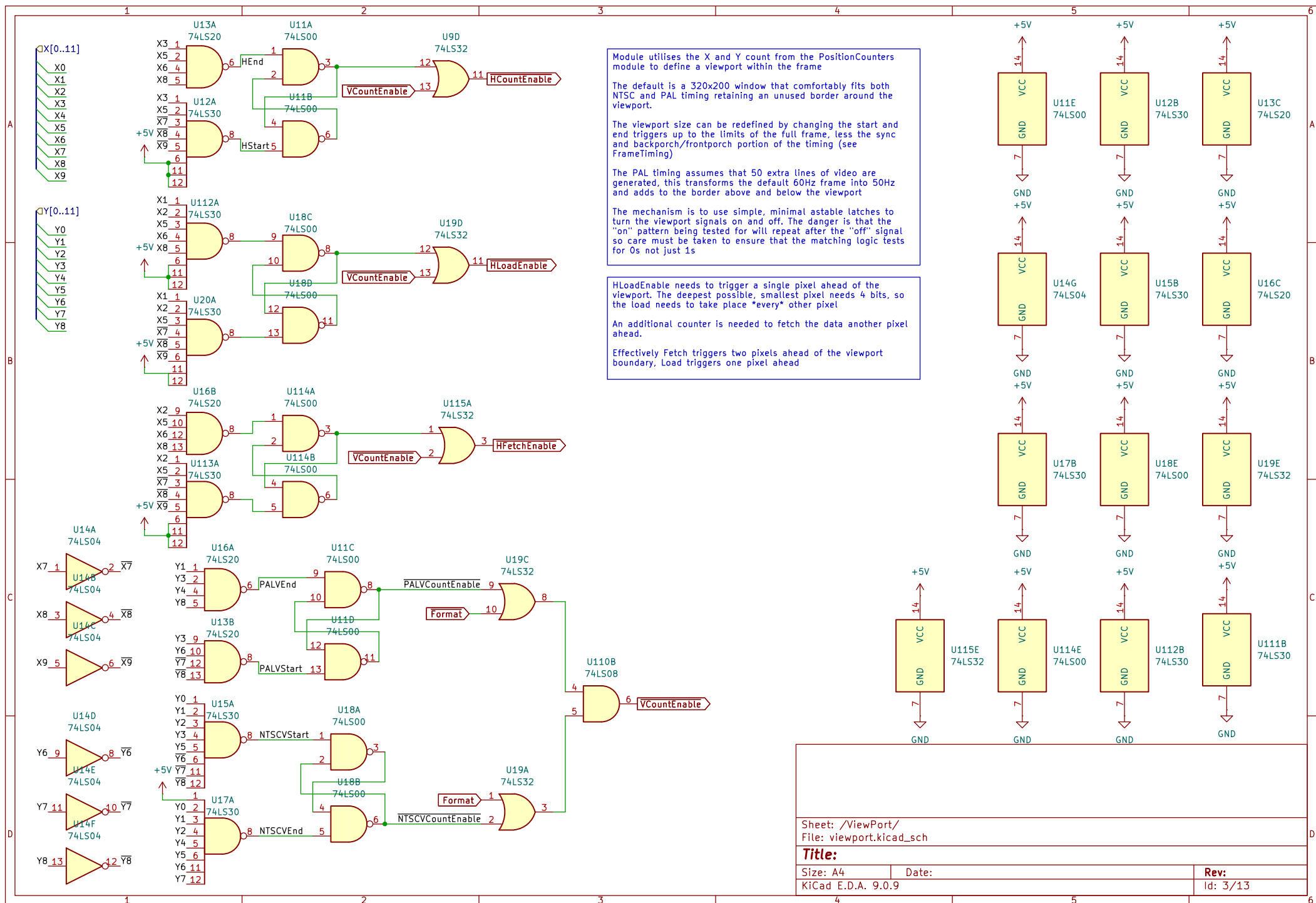
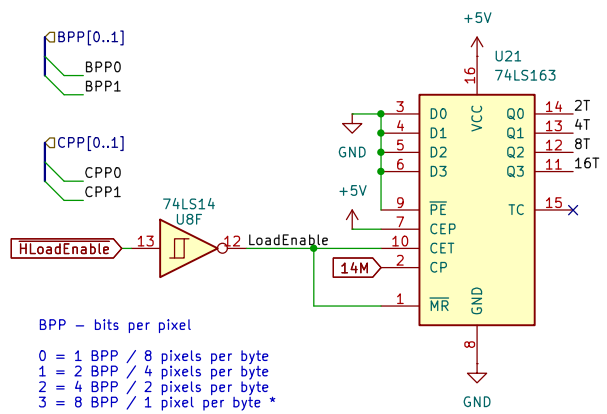










BPP – bits per pixel

0 = 1 BPP / 8 pixels per byte
1 = 2 BPP / 4 pixels per byte
2 = 4 BPP / 2 pixels per byte
3 = 8 BPP / 1 pixel per byte *

* not used

CPP – clocks per pixel

0 = 2 ticks per pixel (HRES = 320)
1 = 4 ticks per pixel (HRES = 160)
2 = 8 ticks per pixel (HRES = 80) *
3 = 16 ticks per pixel (HRES = 40) *

* not used

The BitmapDataCounter module defines when the BitmapPixelData module needs to load a fresh byte of data from the bitmap data bus

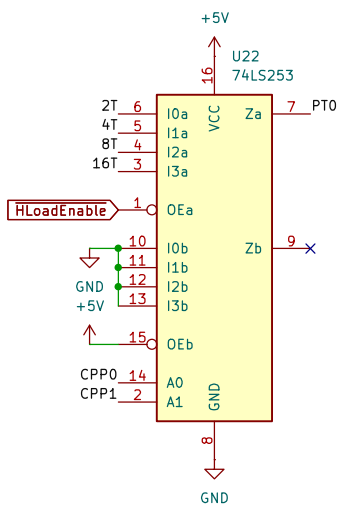
To do calculate when to load the module needs to run a separate counter clocked off the same source as the main pixel counter

The ViewPort module contains a trigger to start the counter out of phase with the data boundary so the data is correctly loaded in advance. Note this does not trigger the memory access, that requires another trigger event in another phase so that the data bus can be prepared before the load action takes place

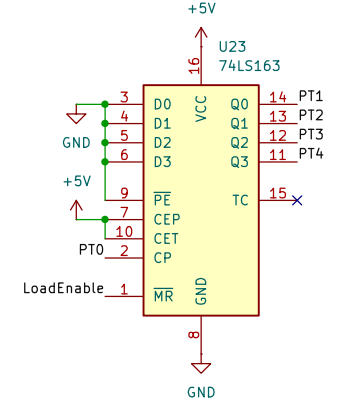
The number of clock ticks per pixel and bits per pixel both impact how frequently the load signal fires. In this simplified design the faster timing options are not used but can be implemented from this design if needed

U21 counts raw clock ticks and acts as a clock divider to reduce the clock by 2, 4, 8 or 16

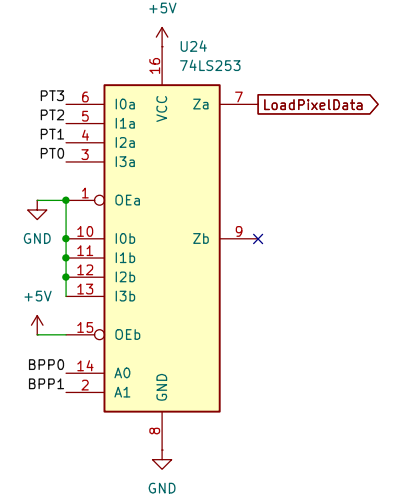
HLoadEnable is a continuous signal, the duration of which is the width of the viewport



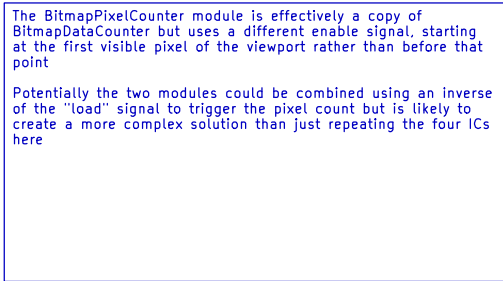
U22 selects the effective clock division based on the clocks per pixel input

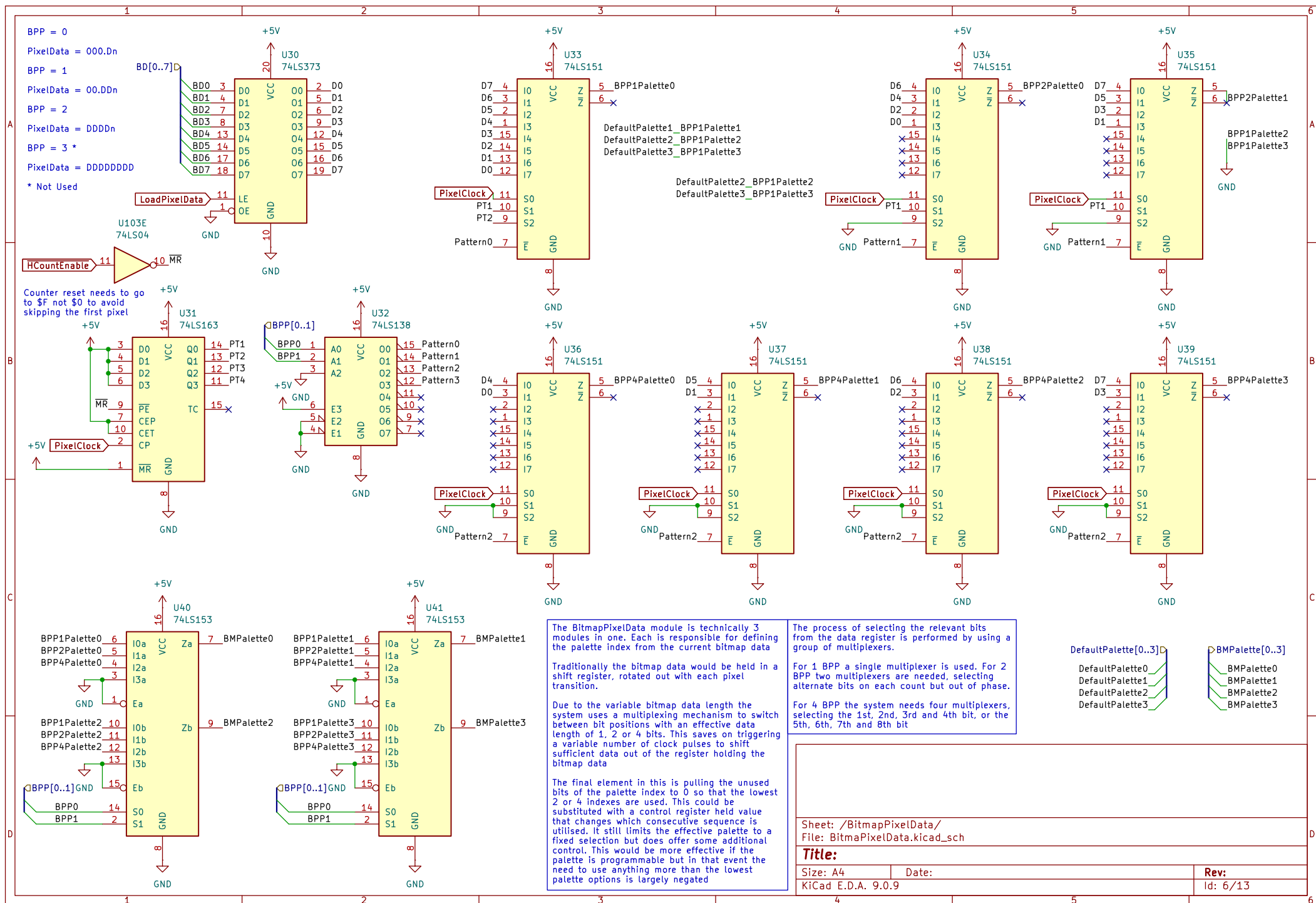


U23 is the effective counter in this circuit, counting pixels

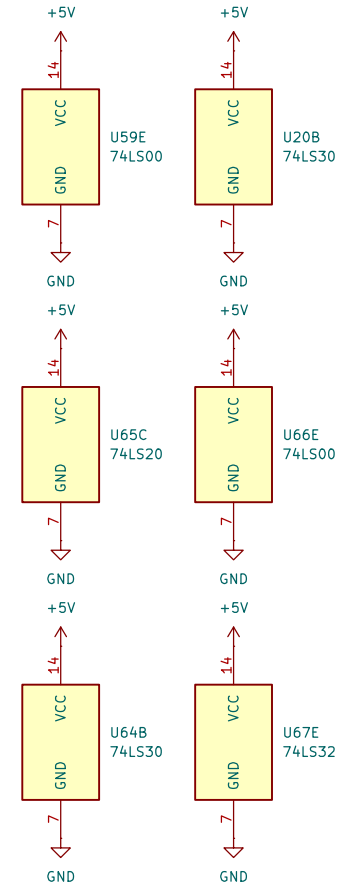
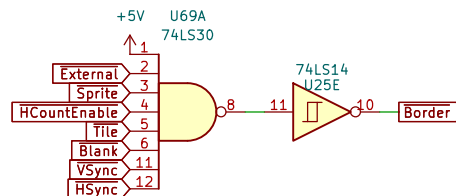


The final stage, U24 selects which pixel count is the effective load pixel data trigger based on the bits per pixel. Note the clock order is reversed as 1 bit per pixel needs more clocks before loading data than 4 bits per pixel





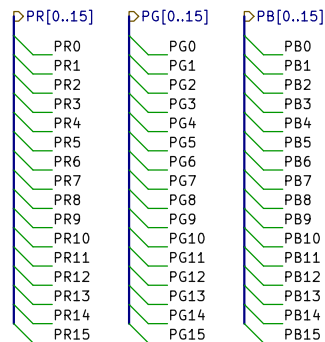
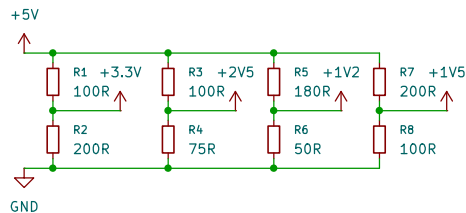
Other select signals are provided by their modules but the encoder uses a priority mechanism to ensure that nothing can override Sync and Porch phases and as such guarantees the frame timing remains stable



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16 Colour Palette

0 = black 0/0/0
1 = white 3.3/3.3/3.3
2 = red 3.3/0/0
3 = blue 0/3.3/0
4 = green 0/0/3.3
5 = yellow = 3.3/3.3/0
6 = magenta = 3.3/0/3.3
7 = cyan = 0/2.2/2.2
8 = d.grey = 1.1/1.1/1.1
9 = l.grey = 2.2/2.2/2.2
10 = m.red = 1.6/0/0
11 = m.blue = 0/0/1.6
12 = m.green = 0/1.6/0
13 = orange = 3.3/1.6/0
14 = violet = 2.2/0/2.2
15 = teal = 0/1.1/1.1



Sheet: /PaletteGenerator/
File: PaletteGenerator.kicad_sch

Title:

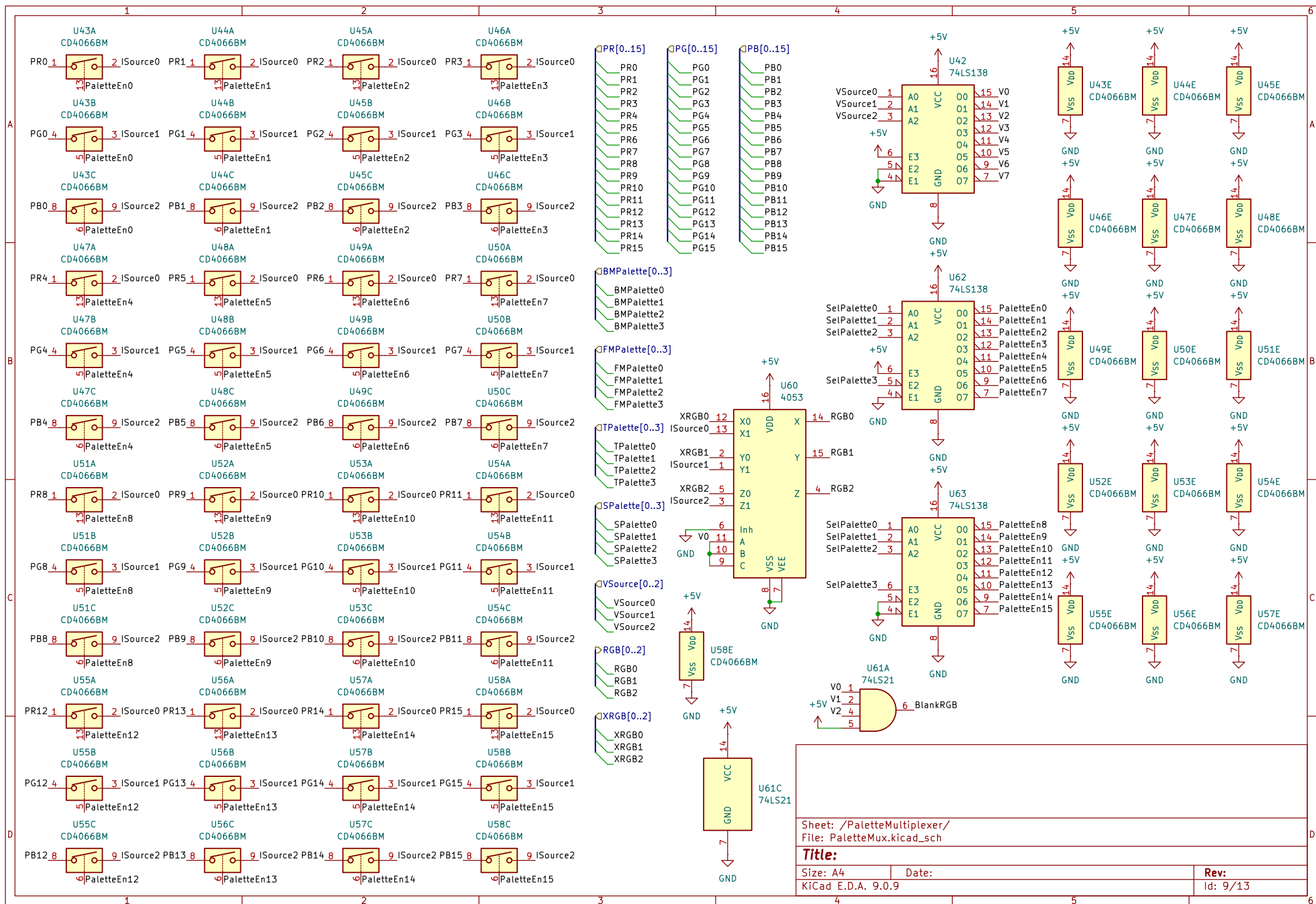
Size: A4

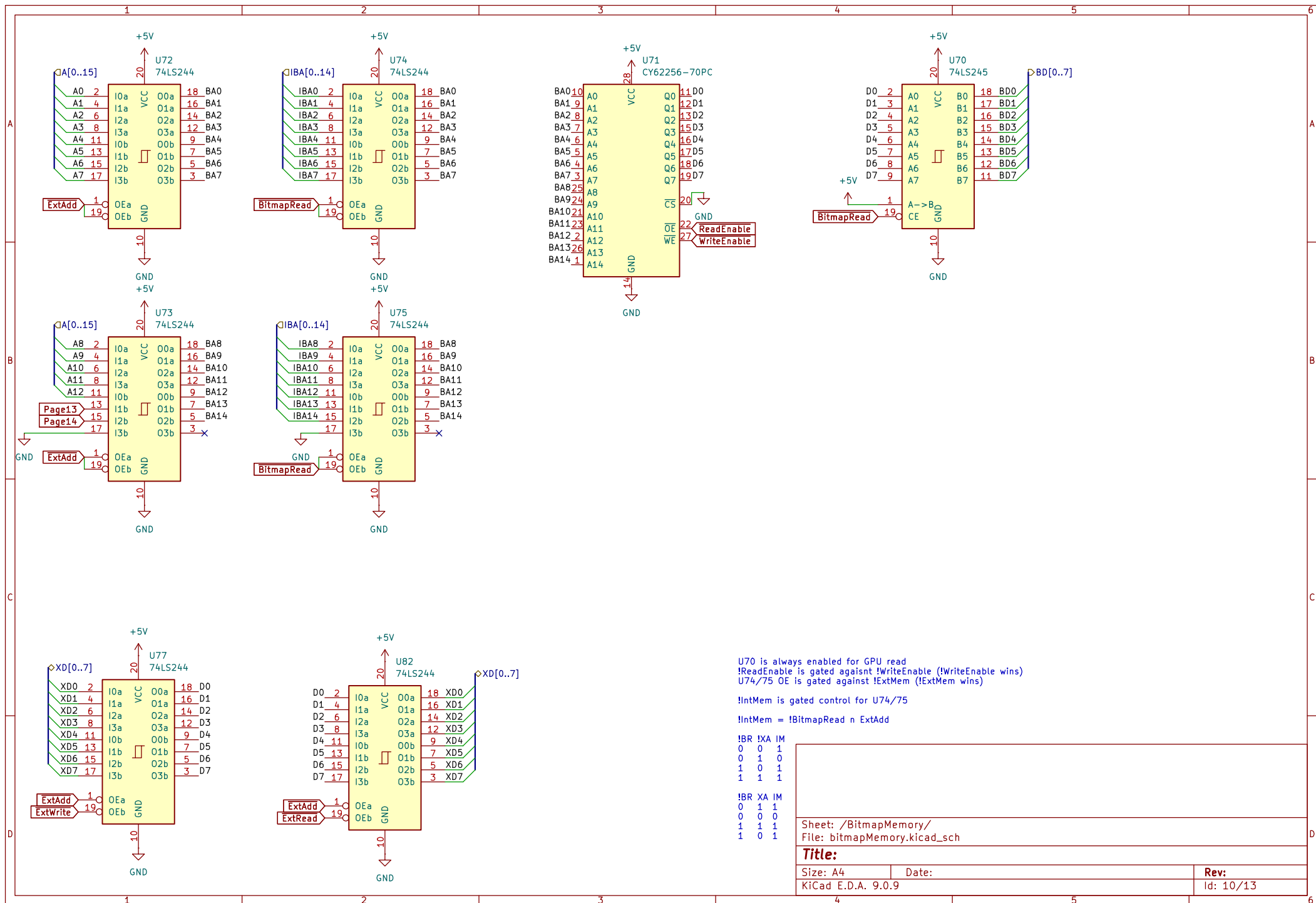
Date:

Rev:

KiCad E.D.A. 9.0.9

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U70 is always enabled for GPU read
 !ReadEnable is gated against !WriteEnable (!WriteEnable wins)
 U74/75 OE is gated against !ExtMem (!ExtMem wins)

!IntMem is gated control for U74/75

!IntMem = !BitmapRead n ExtAdd

!BR !XA IM
 0 0 1
 0 1 0
 1 0 1
 1 1 1

!BR XA IM
 0 1 1
 0 0 0
 1 1 1
 1 0 1

Sheet: /BitmapMemory/
 File: bitmapMemory.kicad_sch

Title:

Size: A4

Date:

Rev:

KiCad E.D.A. 9.0.9

Id: 10/13

